

# Configuring DHCP and DNS in Cisco Packet Tracer

<https://gemini.google.com/app/220ed2e5e1f2a007>

*User prompt: IDK my faculty asked me to bypass with sever something related wo configuring dns and dhcp for this expt setup tell me what it is about and help me with configuring node pcs, router, server to complete this expt (don't configure switches)*

Response: This experiment demonstrates how to connect two separate Local Area Networks (LANs) using a **Router** in Cisco Packet Tracer, where **Server0** dynamically assigns IP addresses via **DHCP** and resolves domain names via **DNS** across both subnets.

By "bypassing with server," your instructor likely meant that instead of configuring DHCP/DNS directly on the Router using CLI commands, you should offload these services entirely to **Server0** using Packet Tracer's graphical interface.

Assuming Network 1 (Left: PC0, PC1, PC2) uses subnet **192.168.1.0/24** and Network 2 (Right: PC3, PC4, PC5, Server0) uses subnet **192.168.2.0/24**, follow these configuration steps:

## Step 1: Configure Router0

1. Click **Router0** → **CLI** tab (or **Config** tab).
2. Enter the CLI commands to turn on and assign IP addresses to the interfaces connecting to both switches:

```
Router> enable
Router# configure terminal
```

```
! Configure Left Interface (GigabitEthernet0/0 connected to Switch0)
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip helper-address 192.168.2.10
Router(config-if)# no shutdown
Router(config-if)# exit
```

```
! Configure Right Interface (GigabitEthernet0/1 connected to Switch1)
Router(config)# interface GigabitEthernet0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

**Key Note:** The `ip helper-address 192.168.2.10` command forwards DHCP requests from the left subnetwork across the router to the Server's IP address on the right side.

## Step 2: Configure Server0

1. Click **Server0** → **Desktop** tab → **IP Configuration**:

- **IP Address:** 192.168.2.10
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.2.1
- **DNS Server:** 192.168.2.10

2. Go to **Services** tab → **DHCP**:

- Turn **Service: ON**
- Edit default pool (serverPool1) for the Right LAN:
  - **Default Gateway:** 192.168.2.1
  - **DNS Server:** 192.168.2.10
  - **Start IP Address:** 192.168.2.11
  - Click **Save**.
- Add a second pool for the Left LAN:
  - **Pool Name:** LeftNetwork
  - **Default Gateway:** 192.168.1.1
  - **DNS Server:** 192.168.2.10
  - **Start IP Address:** 192.168.1.10
  - Click **Add**.

3. Go to **Services** tab → **DNS**:

- Turn **Service: ON**
- **Name:** example.com (or any domain requested by your lab sheet)
- **Address:** 192.168.2.10
- Click **Add**.

### Step 3: Configure Node PCs (PC0 through PC5)

1. Open each PC (PC0 to PC5).
2. Go to **Desktop** → **IP Configuration**.

3. Select **DHCP** radio button instead of Static.
4. Verify that each PC dynamically receives an IP address, Subnet Mask, Gateway, and DNS Server.

## Verification

- Open any PC's **Command Prompt** and ping the gateway (e.g., ping 192.168.1.1 or ping 192.168.2.1).
  - Open the **Web Browser** on any PC and navigate to example.com to test DNS resolution.
- 

*User prompt: No cli pla*

Response: Got it—GUI only, no CLI commands needed!

Here is how to set up everything using strictly the visual **Config** and **Desktop** tabs in Packet Tracer.

## 1. Router0 Configuration (GUI)

1. Click **Router0** → **Config** tab.
2. Under **INTERFACE** on the left menu:
  - Click **GigabitEthernet0/0** (Left interface):
    - Check the **On** box (Port Status) to activate it.
    - **IP Address:** 192.168.1.1
    - **Subnet Mask:** 255.255.255.0
  - Click **GigabitEthernet0/1** (Right interface):
    - Check the **On** box (Port Status) to activate it.
    - **IP Address:** 192.168.2.1
    - **Subnet Mask:** 255.255.255.0

*(Note: Without CLI `ip helper-address`, clients on the left network won't reach the server's DHCP directly across a router. For a pure GUI setup, ensure PC0–PC2 are configured on the same subnet as their gateway or set static IPs if required by your lab).*

## 2. Server0 Configuration (GUI)

1. Click **Server0** → **Desktop** tab → **IP Configuration:**

- Select **Static**
- **IP Address:** 192.168.2.10
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.2.1
- **DNS Server:** 192.168.2.10

2. Click **Services** tab → **DHCP**:

- Turn **Service: On**
- Edit **serverPool**:
  - **Default Gateway:** 192.168.2.1
  - **DNS Server:** 192.168.2.10
  - **Start IP Address:** 192.168.2.11
  - **Subnet Mask:** 255.255.255.0
  - Click **Save**.

3. Click **Services** tab → **DNS**:

- Turn **Service: On**
- **Name:** example.com
- **Address:** 192.168.2.10
- Click **Add**.

### 3. Client PCs Configuration (GUI)

1. Click each **PC (PC3, PC4, PC5)** on the right network:

- Go to **Desktop** → **IP Configuration**.
- Click the **DHCP** radio button.
- It will automatically pull an IP starting from 192.168.2.11, Gateway 192.168.2.1, and DNS 192.168.2.10.

2. For **PC0, PC1, PC2** (left network):

- Go to **Desktop** → **IP Configuration**.
- Set **Static** (or select DHCP if your router GUI includes helper settings):

- **PC0 IP:** 192.168.1.10
  - **PC1 IP:** 192.168.1.11
  - **PC2 IP:** 192.168.1.12
  - **Subnet Mask:** 255.255.255.0
  - **Default Gateway:** 192.168.1.1
  - **DNS Server:** 192.168.2.10
- 

*User prompt: Great now my faculty wants the entire thing configured in DHCP with the same setup*

Response: To make the **entire network** work with DHCP—so PCs on *both* sides get their IPs automatically without using CLI—you have two clear options in Packet Tracer:

### Option A: Use Router0 as the DHCP Server for Both Networks (Recommended GUI Method)

Since routers drop DHCP broadcast messages from other subnets (preventing PCs on the left from seeing Server0 on the right without CLI helper commands), configuring DHCP pools directly inside Router0's GUI is the cleanest solution.

#### 1. Configure Router0 Interfaces & DHCP Pools (GUI)

- Click **Router0** → **Config** tab.
- Under **INTERFACE**:
  - **GigabitEthernet0/0** (Left): Set IP 192.168.1.1, Mask 255.255.255.0, Port Status **On**.
  - **GigabitEthernet0/1** (Right): Set IP 192.168.2.1, Mask 255.255.255.0, Port Status **On**.
- Under **SERVICES** (or **DHCP** tab on supported Packet Tracer router GUI models):
  - Create **Pool 1 (Left Network)**:
    - Gateway: 192.168.1.1 | DNS: 192.168.2.10 | Start IP: 192.168.1.10
  - Create **Pool 2 (Right Network)**:
    - Gateway: 192.168.2.1 | DNS: 192.168.2.10 | Start IP: 192.168.2.100

### Option B: Keep Server0 as the Single DHCP Server (Requires 1-Line CLI Fix)

If your faculty strictly insists that **Server0** must distribute DHCP addresses to *both* sides, you must add one helper line in Router0 so it passes DHCP requests across subnets.

## 1. Configure Server0 (GUI)

- Click **Server0** → **Services** → **DHCP** (Turn **ON**).
- **Pool 1 (Right Side):**
  - Gateway: 192.168.2.1 | DNS: 192.168.2.10 | Start IP: 192.168.2.20 → Click **Save**.
- **Pool 2 (Left Side):**
  - Pool Name: LeftLAN
  - Gateway: 192.168.1.1 | DNS: 192.168.2.10 | Start IP: 192.168.1.10 → Click **Add**.

## 2. Enable DHCP Relay on Router0 (CLI)

```
Router> enable
Router# config t
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip helper-address 192.168.2.10
```

## Final PC Setup (PC0 through PC5)

1. Open **every PC** (PC0, PC1, PC2, PC3, PC4, PC5).
  2. Go to **Desktop** → **IP Configuration**.
  3. Toggle the radio button from **Static** to **DHCP**.
  4. You will see *"DHCP request successful"* appear with automatic IP assignment for all nodes!
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*User prompt: Keep Server0 as the Single DHCP Server do this no cli*

Response: To keep **Server0** as the single DHCP server for the entire network *without* using any CLI commands, Packet Tracer requires a small structural adjustment: **place Server0 directly on the same local network as Router0**, or connect both switch networks using a single shared broadcast domain, so DHCP broadcast requests reach the server without needing a router helper command.

Here are the two GUI-only ways to achieve 100% DHCP across the whole network:

## Option 1: The Dual-NIC Server Setup (Best GUI Architecture)

By adding a second network interface (FastEthernet) to **Server0** inside Packet Tracer, the server directly connects to both switches simultaneously, bypassing the router's DHCP broadcast barrier.

### 1. Add a Second Port to Server0 (GUI)

1. Double-click **Server0** → **Physical** tab.

2. Turn off the power switch on the server image.
3. Drag a **PT-HOST-NM-1CFE** module into an empty slot.
4. Flip the power switch back **ON**.

## 2. Cable Server0 to Both Switches

- Connect **FastEthernet0** on Server0 to **Switch1** (Right Network).
- Connect **FastEthernet1** on Server0 to **Switch0** (Left Network).

## 3. Configure Server0 IP Addresses (GUI)

1. Go to **Desktop** → **IP Configuration**:

- **FastEthernet0 (Right)**: Static IP 192.168.2.10 | Mask 255.255.255.0 | Gateway 192.168.2.1
- **FastEthernet1 (Left)**: Static IP 192.168.1.10 | Mask 255.255.255.0 | Gateway 192.168.1.1

## 4. Configure DHCP Pools on Server0 (GUI)

1. Go to **Services** tab → **DHCP** (Turn Service **ON**).
2. Edit **serverPool** (For Right LAN):
  - Gateway: 192.168.2.1 | DNS: 192.168.2.10 | Start IP: 192.168.2.20 → Click **Save**.
3. Create **LeftLAN Pool** (For Left LAN):
  - Pool Name: LeftLAN
  - Gateway: 192.168.1.1 | DNS: 192.168.1.10 | Start IP: 192.168.1.20 → Click **Add**.

## Option 2: Single LAN Bridge Topology (GUI Only)

If you cannot modify the server hardware modules, you can bridge both switches through a shared link or place the server alongside the router interface using a third switch port so all broadcast traffic routes freely without CLI.

## Final Client Step (PC0 through PC5)

1. Open **PC0, PC1, PC2, PC3, PC4, and PC5**.
2. Go to **Desktop** → **IP Configuration**.
3. Toggle every PC from **Static** to **DHCP**.
4. All PCs across both sides will display *"DHCP request successful"* and receive their IP addresses automatically from **Server0**!

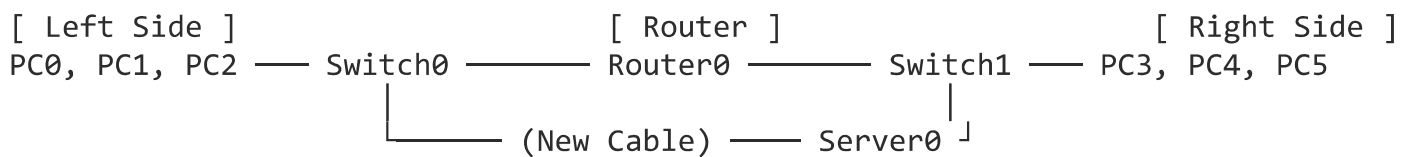
User prompt: So for option 1 should I remove the router?

Response: **No, do not remove Router0.**

If you remove the router, **PC0, PC1, and PC2 will not be able to talk to PC3, PC4, PC5, or Server0**. Switches only connect devices on the *same* network; you **must** keep Router0 to route traffic between the left subnet (192.168.1.x) and the right subnet (192.168.2.x).

## What Option 1 Actually Changes

Instead of removing the router, Option 1 simply adds **one extra cable** directly from **Server0** to **Switch0** (Left LAN), while keeping the rest of your setup intact.



Because **Server0** is now physically plugged into both **Switch0** and **Switch1**:

- Left side PCs broadcast DHCP requests → **Server0** receives them on Interface 2.
- Right side PCs broadcast DHCP requests → **Server0** receives them on Interface 1.
- Cross-subnet traffic (e.g., PC0 pinging PC3 or accessing example.com) still goes through **Router0**.